PCB Inspection Report

Timestamp	2026-04-04 16:33:14
Status	DEFECTIVE
Risk Level	HIGH
Failure Probability	70.06%

Detected Defects

#	Label	Confidence	BBox
1	Missing_hole	0.7954	[1192, 280, 1214, 301]
2	Missing_hole	0.6899	[802, 55, 848, 100]
3	Missing_hole	0.6693	[705, 55, 745, 98]
4	Missing_hole	0.649	[532, 274, 553, 297]
5	Missing_hole	0.5945	[108, 198, 131, 222]
6	Missing_hole	0.3384	[598, 56, 639, 98]
7	Missing_hole	0.3038	[954, 704, 973, 722]

Explanation

1) Root Cause Analysis
The presence of seven distinct "Missing_hole" defects across the PCB, with varying but generally high confidence levels (e.g., 0.7954, 0.6899, 0.6693), strongly indicates a fundamental systemic issue within the PCB fabrication drilling process. This is not attributable to post-drilling contamination, reflow, or soldering issues. Likely manufacturing process contributors include:
a) Worn or broken drill bits leading to incomplete penetration or deviation.
b) Incorrect or corrupted drill programming, causing specific drill points to be skipped or improperly targeted.
c) Miscalibration of the drilling machine, resulting in registration errors where the drill bit does not align with the intended hole locations.
d) Incorrect feed/speed rates for the specific laminate material, causing drill bit deflection or premature wear. The presence of lower confidence detections (0.3384, 0.3038) might suggest partially formed holes or severe undersizing rather than complete absence, further pointing to drill tool or parameter issues.
2) Operational Risks
The detection of multiple critical "Missing_hole" defects with a HIGH risk level poses severe operational challenges. If these holes are intended for through-hole component leads, their absence will prevent proper component insertion, leading to immediate open circuits or incorrect component placement. If these are critical vias, the consequence is an immediate open circuit in the signal or power path, rendering the board non-functional. For mechanical mounting holes, the absence compromises board securement and potentially leads to mechanical stress or vibration failures. The high failure probability of 70.06% directly correlates with the certainty that these missing holes will prevent the intended electrical or mechanical functionality.
3) Failure Prediction

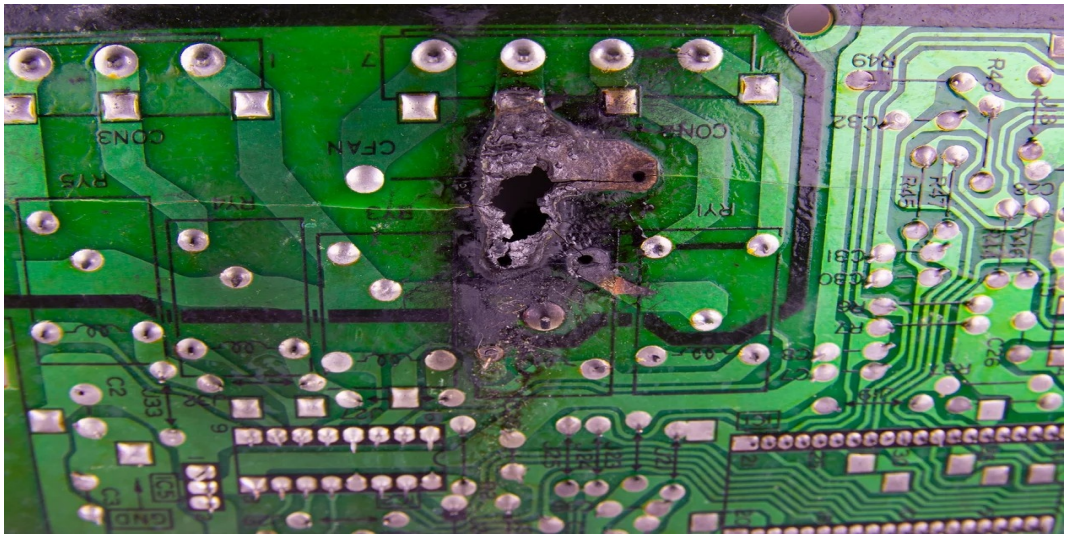
Given the 70.06% failure probability and the nature of "Missing_hole" defects, immediate failure is predicted. Failure will most likely occur during the assembly phase (inability to insert components or mount the board), or during initial power-up and functional testing due to open circuits for critical electrical connections. The board is highly unlikely to pass basic electrical continuity or functional tests. Should components be improperly forced into partially formed holes (as potentially indicated by lower confidence detections), delayed failures such as intermittent connections or early fatigue fractures of component leads under thermal cycling or vibration are probable.
4) Repair Explanation
Repair for multiple "Missing_hole" defects, especially those intended as plated through-holes for critical components, is generally impractical, cost-prohibitive, and would severely compromise the long-term reliability of the PCB. Attempting to re-drill holes on an already laminated and fabricated board risks damaging adjacent traces, creating delamination, or compromising the integrity of surrounding dielectric material. Subsequent plating of re-drilled holes would be challenging to achieve reliably without specialized and expensive processes not typically available in standard rework scenarios. Therefore, given the critical nature and quantity of these defects, these specific boards should be designated as scrap.
5) Engineering Insights
The consistent reporting of "Missing_hole" defects across multiple locations (as indicated by the seven bbox coordinates) demands an immediate and thorough investigation into the PCB fabrication facility's drilling process. The varying confidence levels suggest that the issue might manifest in different degrees, from completely absent holes (e.g., confidence 0.7954) to partially formed or severely constricted ones (e.g., confidence 0.3038). Engineering should request detailed statistical process control (SPC) data for the specific drilling machines, drill bit usage logs, and drill programs for the manufacturing lot. The bbox coordinates are essential for pinpointing the exact locations on the Gerber files and drill programs to identify any potential programming errors or specific machine spindle failures. AOI calibration should be verified to ensure accurate detection across the entire board surface, although the high confidence values suggest robust detection of physical defects.

Repair Suggestions

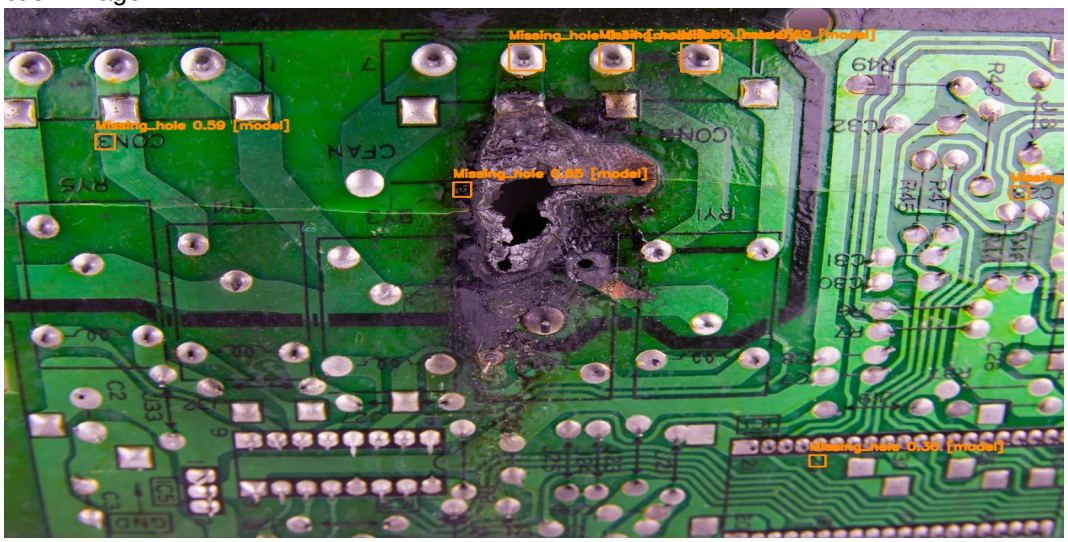
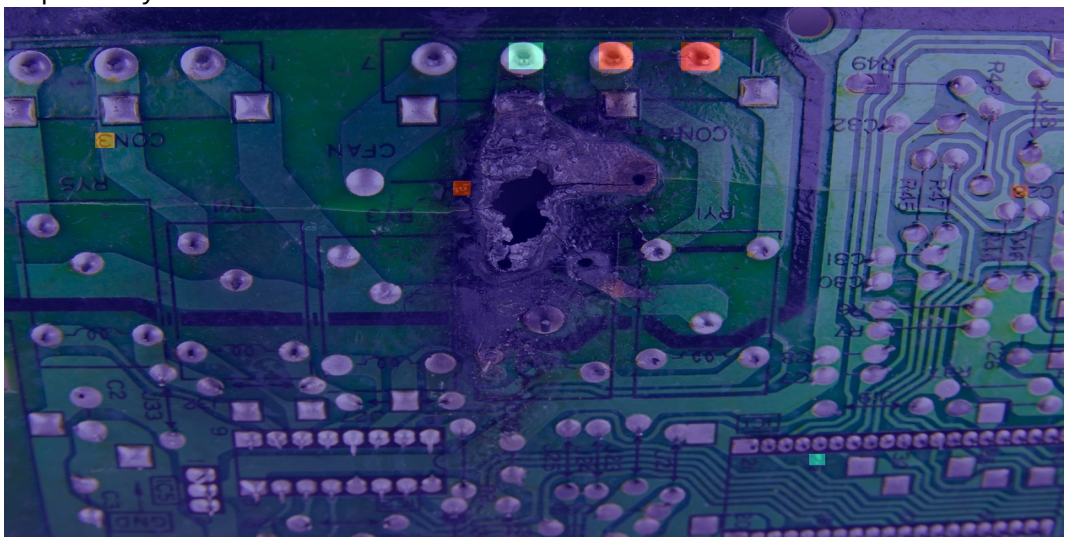
- Missing_hole: Replace the missing component and verify orientation and BOM value before reflow.

Inspection Images

Original Image



Detection Image



Heatmap Overlay